

Simultaneous Diophantine Approximation for Compact Discrete Gaussian Sampling

Ke Ma¹, Jiabo Wang², Shanxiang Lyu¹, Junzuo Lai¹ and Zsolt Lángi³

Abstract

Discrete Gaussian Sampling (DGS) over the integers—also known as integer Gaussian sampling—is used to generate integer values that statistically follow the discrete Gaussian distribution and plays a central role in lattice-based cryptography. Among existing approaches for integer DGS, the cumulative distribution table (CDT) method is widely adopted. However, CDT sampling typically incurs substantial storage costs due to the need to store high-precision fixed-point probability tables, where a precision of k bits is required to achieve a statistical distance of 2^{-k} from the ideal distribution.

In this work, we propose a more compact representation of CDT based on Simultaneous Diophantine Approximation (SDA). Instead of storing fixed-point values, our method expresses the probabilities in the CDT as a sequence of rational numbers with a common denominator. With parameter selection guided by SDA, this compact fractional representation enables reducing data width while maintaining the same level of statistical accuracy. Our SDA-CDT construction offers clear advantages in both computation speed and storage compared to classical CDT implementations. For example, in Frodo-1344, our sampler achieves a 19.97% increase in speed (from 12.10 million to 14.51 million samples per second) and a 3.85% reduction in memory usage (from 104 bits to 100 bits). Similarly, in Frodo-976, we observe a 10.88% speedup and a 21.60% decrease in memory cost. In addition, our design eliminates floating-point arithmetic and supports a fully constant-time online sampling procedure, which ensures resistance to timing side-channel attacks without compromising performance.

Index Terms

Discrete Gaussian Sampling, Diophantine Approximation, Lattice-Based Cryptography.

I. INTRODUCTION

Lattice-based cryptography is currently the most promising candidate for post-quantum standardization, as evidenced by the inclusion of lattice-based schemes among all four finalists in NIST’s post-quantum cryptography competition [1]. The security of these schemes typically relies on the worst-case hardness of fundamental lattice problems such as Learning With Errors (LWE) [2] and Short Integer Solutions (SIS) [3]. The discrete Gaussian distribution is a central tool in such constructions. In particular, the discrete Gaussian distribution underpins several core functionalities: it provides entropy for hardness guarantees through the leftover hash lemma, ensures suitable noise in LWE-based encryption, and enables trapdoor sampling for signatures via the GPV framework [4]. However, sampling from an ideal discrete Gaussian is infeasible in practice. Consequently, implementations target distributions that are statistically close to the ideal one, up to a negligible distance.

One widely-used approach to DGS is via the Cumulative Distribution Table (CDT) method, where precomputed cumulative probabilities guide inverse transform sampling. As the entries of the CDT are generally irrational, they

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must be approximated numerically. Since floating-point representations incur performance and hardware inefficiencies, fixed-point representations are preferred for speed and compatibility but require high bit precision, leading to large entries for the table.

A more compact alternative is to approximate CDT entries by rational numbers with small denominators. Existing techniques focus on bit-wise truncation. Bit-wise truncation approximates x via its k -bit binary expansion:

$$x_k := \frac{\lfloor 2^k x \rfloor}{2^k},$$

with denominator $q = 2^k$ and approximation error bounded by

$$|x - x_k| \leq \frac{1}{2^k} = \frac{1}{q}.$$

Thus, the error scales as $O(1/q)$, and the approximates are restricted to dyadic rationals. We observe that the approximation can be made much more efficient by using Diophantine approximation. It seeks to minimize approximation error for a given denominator size. Classical results, such as Dirichlet’s theorem [5], state that for any $x \in \mathbb{R}$ and $Q > 0$, there exists a rational number $\frac{p}{q}$ with $1 \leq q \leq Q$ such that

$$\left| x - \frac{p}{q} \right| < \frac{1}{qQ}.$$

Setting $Q = q$ yields

$$\left| x - \frac{p}{q} \right| < \frac{1}{q^2},$$

which provides an error scaling of $O(1/q^2)$ —asymptotically superior to bit-wise truncation. Stronger bounds, such as those from Hurwitz’s theorem, can be obtained via continued fraction expansions.

Despite the superiority of Diophantine approximation, most existing CDT-based DGS implementations rely on bit-wise truncation [6], [7]. In this work, we advocate for and implement a CDT representation grounded in Simultaneous Diophantine Approximation (SDA), achieving improved compression and statistical accuracy without compromising constant-time performance.

A. Our Contributions

We introduce a novel CDT construction based on simultaneous Diophantine approximation (SDA). Our contributions span three core aspects, summarized as follows:

- **SVP-Based SDA Construction:** We refine the classical LLL-based SDA by employing a stronger Shortest Vector Problem (SVP) oracle to achieve significantly tighter common denominators. Specifically, this reduces the worst-case bound on the shared denominator q by a factor of $2^{(n+1)/4}$, a substantial theoretical improvement. Although SVP solvers are computationally more expensive than LLL, they remain practical for low-dimensional settings, e.g. to approximate no more than 20 probabilistic numbers in a cumulative distribution table. Moreover, the existing LLL-based SDA only outperforms naive fixed-point truncation when the number of real numbers to be approximated is below a certain threshold, whereas the proposed SVP-based approach is not subject to this limitation.
- **SDA-Enabled CDT Sampler:** Using our SVP-based SDA, we construct a CDT table with rational entries sharing a small common denominator $q < 2^k$, ensuring a statistical distance of at most 2^{-k} from the target distribution. All rational approximations are computed offline, amortizing the cost across deployments. Benchmarks on Frodo and Falcon show that our method reduces memory usage by up to **21.6%** (e.g., from 162 bits to 127 bits in Frodo-976), while achieving up to **19.97%** faster sampling speed (e.g., from 12.10M/s

to 14.51M/s in Frodo-1344). Moreover, randomness cost per sample is reduced by up to **12.5%** in select Frodo parameters.

- **Constant-Time Online Sampling:** The resulting sampler operates in strictly constant time, making it resistant to timing side-channel attacks. We provide a full constant-time implementation for lattice-based schemes including Falcon and Frodo. For typical table sizes (e.g., 30 entries in Frodo and Falcon), the design supports efficient and secure lookup logic without performance degradation. We also give the security analysis of the proposed sampler based on Rényi divergence and statistical distance.

B. Related Work

The design of discrete Gaussian samplers (DGS) has attracted considerable attention in the cryptographic literature, leading to a variety of techniques with distinct trade-offs.

Rejection sampling methods [4], though conceptually straightforward, require high-precision evaluations of exponential functions and exhibit variable runtime behavior, which may expose implementations to timing side-channel attacks. Binary sampling techniques [8] improve efficiency but rely on the cumulative distribution table (CDT) as a base sampler. Our experiments indicate that CDT operations account for approximately 40% of total sampling time in BLISS-like implementations. Rounded Gaussian sampling [9] achieves constant-time performance by rounding from continuous Gaussians, but introduces statistical distortions that complicate security proofs. Convolution-based samplers [10] allow flexible parameter tuning via distribution composition but at the cost of added implementation complexity.

The CDT method remains central due to its deterministic timing and avoidance of floating-point arithmetic. Nevertheless, as noted by Aguilar-Melchor et al. [11], achieving statistical distance 2^{-k} from the target distribution requires storing probabilities at k -bit precision, often with $k \geq 128$ for cryptographic applications—imposing significant memory overhead. Bai et al. [6] show that for a scheme to attain λ -bit security, the statistical distance must be below $2^{-\lambda}$. Prest [7] proposed adopting alternative distance measures such as Rényi divergence to obtain tighter security bounds, potentially reducing required precision by 20–30% in practical settings.

Several works have optimized CDT sampling. Pöppelmann et al. [12] introduced a hardware implementation, subsequently refined in [13]–[15]. Du et al. [13] reduced comparison operations by 60–70% using hierarchical prefix comparisons. While Du et al. [16] proposed an efficient software CDT sampler, it remained susceptible to timing attacks. This issue was addressed by Khalid et al. [17], who proposed a time-independent variant. Aguilar-Melchor et al. [18] introduced a non-centered CDT construction that enables a flexible time-memory trade-off, improving on the performance of rejection sampling. Aguilar-Melchor and Ricosset [11] further reduced required precision by 25% via Rényi divergence, without compromising security. Fouque et al. [19] proposed using a reverse cumulative distribution table (RCDT)—the suffix sum of probabilities—which reduces memory requirements. Karabulut et al. [20] introduced a constant-time CDT-based sampler leveraging fusion trees, achieving up to 86% reduction in CDT size. Their parameters are compatible with post-quantum digital signature schemes. In follow-up work [21], they demonstrated that parallelized data access can eliminate memory access latency on CPU platforms. These improvements span both algorithmic techniques and hardware accelerations. Nonetheless, the core challenge of CDT sampling remains: storing and accessing high-precision probability tables is inherently memory-intensive and difficult to optimize without compromising statistical accuracy or security guarantees.

The remainder of the paper is structured as follows: Section II presents mathematical preliminaries. Section III introduces our Diophantine approximation framework. Section IV describes the construction of the SDA-CDT sampler. Section V discusses applications to Frodo [22] and Falcon [19], [23], along with comparative evaluation. Section VI concludes with performance analysis and future research directions.

II. PRELIMINARIES

We write $\mathbb{R}$ and $\mathbb{Z}$ for the real numbers and integers. If S is countable and $f: S \rightarrow \mathbb{R}$, then $f(S) = \sum_{x \in S} f(x)$. Let $\text{Supp}(\mathcal{D})$ be the support of a distribution $\mathcal{D}$, and let U_S be the uniform distribution on a finite set S . We use λ for the security parameter and q_s for the number of queries an attacker may make to the secret-key oracle. $\mathbf{I}_n$ denotes an $n \times n$ identity matrix.

A. Measure

1) *Statistical Distance*: Statistical Distance (SD) Δ of two discrete probability distributions P and Q with the same support S is defined as:

$$\Delta(\mathcal{P}, \mathcal{Q}) := \frac{1}{2} \sum_{x \in S} |\mathcal{P}(x) - \mathcal{Q}(x)|.$$

2) *Rényi Divergence*: We recall the definition of the Rényi divergence (RD). For two discrete distributions $\mathcal{P}$ and $\mathcal{Q}$ such that $S := \text{Supp}(\mathcal{P}) \subseteq \text{Supp}(\mathcal{Q})$ and $a \in (1, +\infty)$, the Rényi divergence defined as:

$$R_a(\mathcal{P} \parallel \mathcal{Q}) := \left(\sum_{x \in S} \frac{\mathcal{P}(x)^a}{\mathcal{Q}(x)^{a-1}} \right)^{\frac{1}{a-1}}.$$

Lemma 1 ([6]): For $a \in [1, +\infty]$. Let $\mathcal{P}$ and $\mathcal{Q}$ two distribution and $\text{Supp}(\mathcal{P}) \subseteq \text{Supp}(\mathcal{Q})$. Then the following properties holds:

- 1) (Multiplicativity) Let $\mathcal{P}$ and $\mathcal{Q}$ two distributions of a pair of variables (X_1, X_2) . Let $\mathcal{P}_i, \mathcal{Q}_i$ denote the marginal distributions of X_i under $\mathcal{P}$ and $\mathcal{Q}$ for $i \in \{1, 2\}$. Then the following holds if X_i are independent

$$R_a(\mathcal{P} \parallel \mathcal{Q}) = R_a(\mathcal{P}_1 \parallel \mathcal{Q}_1) R_a(\mathcal{P}_2 \parallel \mathcal{Q}_2).$$

- 2) (Probability Preservation) Let $A \subseteq \text{Supp}(\mathcal{Q})$ an arbitrary event. If $a \in (1, +\infty)$ then the following holds:

$$\mathcal{Q}(A) \geq \mathcal{P}(A)^{\frac{a}{a-1}} / R_a(\mathcal{P} \parallel \mathcal{Q}).$$

B. Discrete Gaussian Distribution

Definition 2 (Gaussian Function): For any center $c \in \mathbb{R}$ and Gaussian parameter $s \in \mathbb{R}^+$, let

$$\rho_{c,s}(x) := \exp\left(-\frac{\pi(x-c)^2}{s^2}\right).$$

Definition 3 (Discrete Gaussian Distribution): For center $c \in \mathbb{R}$ and parameter $s \in \mathbb{R}^+$, the discrete Gaussian distribution over $\mathbb{Z}$ is defined as:

$$\mathcal{D}_{c,s}(x) := \frac{\rho_{c,s}(x)}{\rho_{c,s}(\mathbb{Z})}, \quad \text{where } \rho_{c,s}(\mathbb{Z}) := \sum_{k \in \mathbb{Z}} \rho_{c,s}(k).$$

This distribution is supported on all integers $x \in \mathbb{Z}$, with probabilities proportional to $\rho_{c,s}(x)$. If c is 0, it can be omitted.

Note that the distribution $\mathcal{D}_{c,s}$ has standard deviation $\sigma = \frac{s}{\sqrt{2\pi}}$.

Definition 4 (Cumulative Distribution Function (CDF)): The cumulative distribution function $\mathcal{T}_{c,s}: \mathbb{Z} \rightarrow [0, 1]$ is defined as:

$$\mathcal{T}_{c,s}(x) := \sum_{\substack{k \in \mathbb{Z} \\ k \leq x}} \mathcal{D}_{c,s}(k).$$

For practical computations, the sum is often truncated to integers within $[c - \tau s, c + \tau s]$ for some $\tau > 0$, ensuring $\mathcal{T}_{c,s}(x)$ approximates the full sum to within negligible error.

C. Lattices

A lattice $\mathcal{L} \subseteq \mathbb{R}^m$ is the set of all integer linear combinations of n linearly independent vectors $\mathbf{b}_1, \dots, \mathbf{b}_n \in \mathbb{R}^m$, where $n \leq m$. Letting $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$, the lattice generated by $\mathbf{B}$ is

$$\mathcal{L}(\mathbf{B}) = \left\{ \sum_{i=1}^n l_i \mathbf{b}_i : l_i \in \mathbb{Z} \right\}.$$

The basis is not unique: different bases can generate the same lattice if they are related by a unimodular matrix (i.e., an integer matrix with determinant ± 1). When the basis is understood from context, we simply write $\mathcal{L}$.

The *determinant* of a lattice $\mathcal{L}$, denoted $\det(\mathcal{L})$, is defined in terms of a basis $\mathbf{B}$ as

$$\det(\mathcal{L}) = \sqrt{\det(\mathbf{B}^\top \mathbf{B})}.$$

If $\mathbf{B}$ is a square matrix (i.e., $n = m$), then the determinant simplifies to $\det(\mathcal{L}) = |\det(\mathbf{B})|$.

Definition 5 (ℓ_p norm): The ℓ_p norm of a vector $\mathbf{x} \in \mathbb{R}^n$ is defined by $\|\mathbf{x}\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ for $1 \leq p < \infty$ and $\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|$ for $p = \infty$.

Remark 6: For $\mathbf{x} \in \mathbb{R}^n$, $\|\mathbf{x}\|_p \leq \|\mathbf{x}\|_2 \leq \sqrt{n} \|\mathbf{x}\|_p$ for $p \geq 2$ and $\frac{1}{\sqrt{n}} \|\mathbf{x}\|_p \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_p$ for $1 \leq p < 2$.

Definition 7 (Successive Minima): With respect to a general norm function, the first successive minimum $\lambda_1(\mathcal{L})$ of a lattice $\mathcal{L}$ is the length of its shortest nonzero vector:

$$\lambda_1(\mathcal{L}) = \min_{\mathbf{x} \in \mathcal{L} \setminus \{0\}} \|\mathbf{x}\|.$$

Definition 8 (Shortest Vector Problem (SVP)): Given a lattice $\mathcal{L}$, the *Shortest Vector Problem (SVP)* asks for a nonzero vector $\mathbf{v} \in \mathcal{L}$ such that $\|\mathbf{v}\|$ is minimized, i.e., $\|\mathbf{v}\| = \lambda_1(\mathcal{L})$.

Although SVP is a fundamental computational challenge in lattice theory, for low to moderate dimensions (e.g., up to 100), exact solution can be obtained by enumeration [24] or sieving [25]. Regarding ℓ_2 -norm SVP, approximation algorithms such as the LLL algorithm and its variants (e.g., BKZ) can approximately find short vectors [26], though they do not guarantee the shortest. In this work, we will adjust the enumeration [24] algorithm to solve ℓ_∞ -norm SVP (See Appendix A). Of independent interests, the proposed adjusted enumeration algorithm can also be applied to solve the minimal rational fraction representation problem in [27], [28].

Theorem 9 (Minkowski's Theorem under the ℓ_2 Norm, [29]): Let $\mathcal{L} \subseteq \mathbb{R}^n$ be a full-rank lattice. Then the ℓ_2 norm of the first successive minimum satisfies:

$$\lambda_1^{(2)}(\mathcal{L}) \leq \sqrt{n} \cdot \det(\mathcal{L})^{1/n}. \quad (1)$$

This work needs to use the ℓ_∞ norm version of Minkowski's theorem.

Theorem 10 (Minkowski's Theorem under the ℓ_∞ Norm): Let $\mathcal{L} \subseteq \mathbb{R}^n$ be a full-rank lattice. Then the first successive minimum with respect to the ℓ_∞ norm satisfies

$$\lambda_1^{(\infty)}(\mathcal{L}) \leq \det(\mathcal{L})^{1/n}. \quad (2)$$

Proof: Let $B_\infty(r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\|_\infty \leq r\}$ denote the closed cube of side length $2r$ centered at the origin. The volume of this cube is:

$$\text{vol}(B_\infty(r)) = (2r)^n. \quad (3)$$

Set $r = \det(\mathcal{L})^{1/n}$. Substituting this value, the volume becomes:

$$\text{vol}(B_\infty(r)) = \left(2 \det(\mathcal{L})^{1/n}\right)^n = 2^n \det(\mathcal{L}). \quad (4)$$

By Minkowski's convex body theorem [29], any compact symmetric convex body $K \subset \mathbb{R}^n$ with $\text{vol}(K) \geq 2^n \det(\mathcal{L})$ contains a nonzero lattice point in $\mathcal{L} \setminus \{0\}$. The cube $B_\infty(r)$ is compact and convex, and its volume equals the critical threshold $2^n \det(\mathcal{L})$. To resolve the borderline case:

- If $B_\infty(r)$ contains no nonzero lattice points, then for some $\epsilon > 0$, the slightly enlarged cube $B_\infty(r + \epsilon)$ would still contain no lattice points. However, $\text{vol}(B_\infty(r + \epsilon)) > 2^n \det(\mathcal{L})$, contradicting Minkowski's theorem.
- Alternatively, the closed cube $B_\infty(r)$ contains all its limit points. If nonzero lattice points approach the boundary as $\epsilon \rightarrow 0$, compactness ensures at least one such point lies within $B_\infty(r)$.

Hence, there exists $\mathbf{v} \in \mathcal{L} \setminus \{0\}$ with $\|\mathbf{v}\|_\infty \leq r$. By definition of the first successive minimum:

$$\lambda_1^{(\infty)}(\mathcal{L}) = \min_{\mathbf{v} \in \mathcal{L} \setminus \{0\}} \|\mathbf{v}\|_\infty \leq \det(\mathcal{L})^{1/n}. \quad (5)$$

□

Compared to the classical Minkowski bound under the ℓ_2 norm, which gives $\lambda_1^{(2)}(\mathcal{L}) \leq \sqrt{n} \cdot \det(\mathcal{L})^{1/n}$, the ℓ_∞ version achieves a strictly tighter bound by a factor of $\sqrt{n}$. This improvement arises from the geometry of the ℓ_∞ ball, which is a cube aligned with the coordinate axes.

D. Diophantine Approximation

Diophantine approximation is a branch of number theory that investigates how well real numbers can be approximated by rational numbers. Formally, given a real number $\alpha \in \mathbb{R}$, the goal is to find integers p and q such that the absolute error $|\alpha - p/q|$ is small for some nonzero integer q . A fundamental question in this area is: how small can this error be made, given constraints on q ?

A classic result in this domain is Dirichlet's approximation theorem [5], which guarantees that for any real number α and any positive integer Q , there exist integers p and q with $1 \leq q \leq Q$ such that

$$\left| \alpha - \frac{p}{q} \right| < \frac{1}{qQ}. \quad (6)$$

This implies that every real number admits infinitely many rational approximations with error less than $1/q^2$, a bound that is optimal up to constant factors for most irrationals.

In many applications, it is necessary to approximate several real numbers simultaneously. This leads to the notion of *simultaneous Diophantine approximation* (SDA). Given a vector $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)^\top \in \mathbb{R}^n$ and a desired error bound $\epsilon > 0$, SDA aims to find integers $q, p_1, \dots, p_n$ such that

$$|q\alpha_i - p_i| \leq \epsilon \quad \text{for all } i = 1, \dots, n, \quad (7)$$

with q as small as possible. This setting naturally arises in cryptography (e.g., lattice-based constructions), signal processing, and the study of Diophantine equations.

An SDA-version of the Dirichlet approximation theorem is given as follows [30]: Given real numbers $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_n)^\top \in \mathbb{R}^n$ and a natural number Q , then there exist integers $q, p_1, \dots, p_n$ with $1 \leq q \leq Q^n$ such that

$$\left| \alpha_i - \frac{p_i}{q} \right| \leq \frac{1}{qQ} \quad \text{for all } i = 1, \dots, n. \quad (8)$$

III. DIOPHANTINE APPROXIMATION FOR COMPACT REPRESENTATION

A trivial method to approximate real numbers with fractions is the fixed-point representation, which is widely used in many designs and implementations.

Definition 11 (k -Bit Fixed-point Representation): The k -bit fixed-point representation of $x \in [0, 1)$ is

$$x_k := \frac{\lfloor 2^k x \rfloor}{2^k}, \quad (9)$$

requiring k bits to store $p = \lfloor 2^k x \rfloor$.

This corresponds to rational approximation with $q = 2^k$, whose approximation error is no larger than 2^{-k} as is provided as below.

Definition 12 (k -Bit Bound): For the k -bit fixed point representation of $\alpha \in \mathbb{R}^n$ with data width $q = 2^k$, the bounds of accuracy is given by

$$|\alpha_i - \lfloor 2^k \alpha_i \rfloor / q| \leq 2^{-k}, \quad \text{for all } i = 1, \dots, n. \quad (10)$$

Definition 13 (k -Bit Compact Representation): For a general approximation of $\alpha \in \mathbb{R}^n$ with integers $p_1, \dots, p_n, q$, it is called k -bit compact if

- 1) the accuracy $|\alpha_i - p_i/q| \leq 2^{-k}$, for all $i = 1, \dots, n$,
- 2) the data width $q < 2^k$.

Put simply, a compact representation achieves no worse accuracy bound by smaller a data width than the fixed-point representation. This work will leverage simultaneous Diophantine approximation to give the compact representation for the probabilities stored in a CDT for discrete Gaussian sampling.

A. Motivating Example

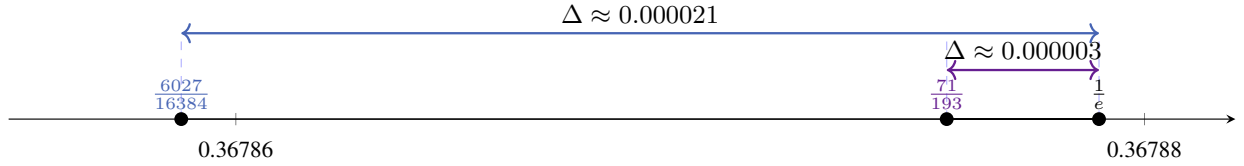


Fig. 1: Comparison of approximations to $\frac{1}{e}$: $\frac{6027}{16384}$ by fixed-point representation vs. $\frac{71}{193}$ by SDA.

We demonstrate SDA for transcendental number $1/e$ to achieve approximation accuracy of 2^{-14} . In comparison with fixed-point representation which requires a 14-bit denominator q , we will show we can achieve a compact representation using SDA. The step-by-step procedures and parameter selection will be illustrated in Theorem 16 in the sequel.

To begin with, we construct a basis $\mathbf{B}$ for a full-rank lattice with $\epsilon = 0.001$ as

$$\mathbf{B} = \begin{bmatrix} \epsilon^{-2} & -\epsilon^{-2}/e \\ 0 & 1 \end{bmatrix}. \quad (11)$$

The shortest vector can be found by SVP as $\mathbf{b} = \mathbf{B}\mathbf{x}$ with the coefficients being

$$\mathbf{x} = \begin{bmatrix} 71 \\ 193 \end{bmatrix}. \quad (12)$$

Therefore, the approximation of $1/e$ by SDA is $71/193$. Compared with fixed-point representation, SDA achieves even better accuracy while giving a favorably more compact representation.

$$\text{14-bit Fixed-point: } \frac{1}{e} \approx \frac{6027}{16384}, \quad \left| \frac{1}{e} - \frac{6027}{16384} \right| \approx 2^{-15}$$

$$\text{14-bit Compact: } \frac{1}{e} \approx \frac{71}{193}, \quad \left| \frac{1}{e} - \frac{71}{193} \right| \approx 2^{-18}$$

B. LLL-Based SDA

An existing approach to solve SDA is based on the Lenstra–Lenstra–Lovász (LLL) lattice basis reduction proposed in [31].

Theorem 14 (LLL-Based SDA, Page 525, [31]): There exists an algorithm that, given $n \in \mathbb{Z}^+$, real numbers $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)^\top \in \mathbb{R}^n$, and error ϵ with $0 < \epsilon < 1$, computes integers $q, p_1, \dots, p_n$ such that:

$$|q\alpha_i - p_i| \leq \epsilon \quad \text{for } 1 \leq i \leq n, \quad (13)$$

$$1 \leq q \leq 2^{n(n+1)/4} \epsilon^{-n}. \quad (14)$$

Note that the original statement in [31] concerns rational vectors α ; here, we have extended it to real vectors α to align with the formulation of the SDA problem. Consequently, we do not assert polynomial-time solvability for this generalized version.

Proposition 15: By making $\epsilon = 2^{-\frac{k-n(n+1)/4}{n}}$, the LLL-based SDA returns a compact approximation of $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)^\top \in \mathbb{R}^n$ if $2^k > q \geq 2^{\frac{n-1}{n}k + \frac{n+1}{4}}$.

Proof: Setting $\epsilon = 2^{-\frac{k-n(n+1)/4}{n}}$, we have $q \leq 2^k$ and

$$\left| \alpha_i - \frac{p_i}{q} \right| \leq q^{-1} \cdot 2^{-\frac{k-n(n+1)/4}{n}}. \quad (15)$$

The compactness condition requires

$$q^{-1} \cdot 2^{-\frac{k-n(n+1)/4}{n}} \leq 2^{-k}, \quad (16)$$

which is equivalent to

$$q \geq 2^{\frac{n-1}{n}k + \frac{n+1}{4}}. \quad (17)$$

□

We also conclude that LLL-based SDA cannot give a compact representation when $n > 2\sqrt{k} - 1/2$ which is derived from

$$2^k < 2^{\frac{n-1}{n}k + \frac{n+1}{4}}.$$

C. SVP-Based SDA

To tighten the bound in Eq. (14), a straightforward consideration is to employ an SVP solver, which produces the shortest nonzero lattice vector at the cost of increased computational complexity.

Theorem 16 (SVP-Based SDA): Given $n \in \mathbb{Z}^+$, real numbers $\alpha = (\alpha_1, \dots, \alpha_n)^\top \in \mathbb{R}^n$, and error ϵ with $0 < \epsilon < 1$, there exist integers $q, p_1, \dots, p_n$ such that:

$$|q\alpha_i - p_i| \leq \epsilon \quad \text{for } 1 \leq i \leq n, \quad (18)$$

$$1 \leq q \leq \epsilon^{-n}. \quad (19)$$

Proof: We constructs a lattice $\mathcal{L}$ with column basis

$$\mathbf{B} = \begin{bmatrix} \epsilon^{-(n+1)} \mathbf{I}_n & -\epsilon^{-(n+1)} \boldsymbol{\alpha} \\ \mathbf{0} & 1 \end{bmatrix}. \quad (20)$$

The determinant of this lattice is $\det(\mathbf{B}) = \epsilon^{-n(n+1)}$.

Then we employ the ℓ_∞ norm SVP solver to find the shortest vector of $\mathcal{L}(\mathbf{B})$, denoted as $\mathbf{b} = \mathbf{B} \cdot [p_1, p_2, \dots, p_n, q]^\top$. Due to the symmetry of lattices, $\mathbf{b} = -\mathbf{B} \cdot [p_1, p_2, \dots, p_n, q]^\top$ is also the shortest vector. W.l.o.g. we assume that $q \geq 0$. Then for the first n and the last coefficient of $\mathbf{b}$, we have

$$\epsilon^{-(n+1)} |q\alpha_i - p_i| \leq \|\mathbf{b}\|_\infty \leq \det(\mathbf{B})^{1/(n+1)} = \epsilon^{-n}, \quad 1 \leq i \leq n \quad (21)$$

$$\Rightarrow |\alpha_i - p_i/q| \leq \epsilon/q, \quad 1 \leq i \leq n, \quad (22)$$

$$\text{and } q \leq \|\mathbf{b}\|_\infty \leq \det(\mathbf{B})^{1/(n+1)} = \epsilon^{-n}. \quad (23)$$

From $\mathbf{b} \neq \mathbf{0}$ and $\epsilon < 1$ we see that $q \neq 0$. Thus the proof is completed. $\square$

Proposition 17: By making $\epsilon = 2^{-k/n}$, the SVP-based SDA returns a compact approximation of $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)^\top \in \mathbb{R}^n$ if $2^k > q \geq 2^{-k/n+k}$.

Proof: Letting $\epsilon = 2^{-k/n}$, we have $q \leq 2^k$ and

$$\left| \alpha_i - \frac{p_i}{q} \right| \leq q^{-1} \cdot 2^{-k/n}. \quad (24)$$

The compactness condition requires

$$q^{-1} \cdot 2^{-k/n} \leq 2^{-k}, \quad (25)$$

which is equivalent to

$$q \geq 2^{-k/n+k}. \quad (26)$$

$\square$

Compared with the LLL-based approach, the advantages of the SVP-based are two-fold. Firstly, unlike the LLL-based SDA which is not able to give a compact representation for $n > 2\sqrt{k} - 1/2$, there is no limitation on the number n of real numbers to be approximated for the SVP-based approach. Secondly, a compact representation requires that the output denominator q satisfies the lower bound in Eq. (26). Hence, the SVP-based bound is strictly better than the LLL-based one. Specifically, for any fixed k and n , we have:

$$q_{\text{LLL}}(k) = q_{\text{SVP}}(k) \cdot 2^{\frac{n+1}{4}} > q_{\text{SVP}}(k). \quad (27)$$

IV. SDA-CDT SAMPLER

This section introduces the standard CDT sampler for comparison, and then presents our SDA-CDT sampler, whose CDT representation is compact.

A. Standard CDT Sampler

We firstly present the standard fixed-point CDT sampler, Algorithms 1 and 2 for reference. In line 4 of Algorithm 1, $\ll$ denotes the bitwise left-shift operator, i.e., for any integer x and non-negative integer k , we define $x \ll k :=$

Algorithm 1: Standard CDT Offline Phase

Input: Centered cumulative distribution table $\mathcal{T}_{c,s}$, precision k
Output: Precomputed table $\mathbf{CDT}[\cdot]$

```

1 Initialize empty table  $\mathbf{CDT}$ 
2  $\mathbf{CDT}[\lceil c - \tau s \rceil - 1] = 0$ 
3 for  $i \leftarrow \lceil c - \tau s \rceil$  to  $\lceil c + \tau s \rceil$  do
4    $\mathbf{CDT}[i] = \mathbf{CDT}[i - 1] + \lfloor (\mathcal{T}_{c,s}(i) - \mathcal{T}_{c,s}(i - 1)) \ll k \rfloor$ 
5 end
6 return  $\mathbf{CDT}$ 
```

Algorithm 2: Standard CDT Online Phase

Input: Precomputed $\mathbf{CDT}[\cdot]$, precision k
Output: Sample $x \sim \mathcal{D}_{c,s}$

```

1  $u \leftarrow \text{UniformBits}(k)$ 
2  $x = \lceil c - \tau s \rceil - 1$ 
3 for  $i \leftarrow \lceil c - \tau s \rceil$  to  $\lceil c + \tau s \rceil$  do
4    $x \leftarrow x + \langle u < \mathbf{CDT}[i] \rangle$ 
5 end
6 return  $x$ 
```

$x \cdot 2^k$. In line 1 of Algorithm 2, $\text{UniformBits}(k)$ returns a uniformly random integer from $\{0, 1, \dots, 2^k - 1\}$ using k bits of uniform randomness. In addition, $\langle E \rangle$ denotes the truth indicator for a logical expression E :

$$\langle E \rangle = \begin{cases} 1 & \text{if } E \text{ holds true} \\ 0 & \text{otherwise} \end{cases}$$

B. SDA-CDT Sampler

The proposed SDA-CDT sampler proceeds with an offline phase and an online phase as in Algorithms 4 and 5. The offline phase employs the SVP-based SDA (Algorithm 3) to compute the rational approximation for the target probability mass function and returns the cumulative distribution table as **SDAT**. Similar to the standard CDT sampler, the online phase involves the generation of random integers, comparison of integers and the execution of addition operations.

The SVP solver in line 7 of Algorithm 3 provides the shortest vector of the input lattice in terms of the ℓ_∞ norm. Pseudocodes of this algorithm are provided in Appendix A. The computation complexity of this SVP solver can be circumvented effectively by situating it within the offline execution phase. Moreover, the parallel computing capability of **SDAT** enables its execution on FPGA or GPU platforms, thereby enhancing the computational efficiency of the offline phase. However, this particular aspect is not within the scope of this paper.

In line 2 of Algorithm 5, $\text{UniformBnd}(m)$ samples uniformly at random from $\{0, 1, \dots, m-1\}$. Unlike $\text{UniformBits}(k)$, this variant ensures unbiased uniformity over an arbitrary bound m , typically used when selecting integer candidates such as lattice coset representatives or precomputed offsets.

C. Security

We consider the k -bit compact approximation $\mathcal{D}_\delta$ of the discrete Gaussian distribution $\mathcal{D}$ using the SDA method. The approximation satisfies $\mathcal{D}_\delta(x_i) = p_i/q$ for x_i in the support $S = \{\lceil c - \tau s \rceil, \lceil c - \tau s \rceil + 1, \dots, \lceil c + \tau s \rceil\}$, where $\{p_i\}$ and q are integers from SDA with guaranteed bound $|q\mathcal{D}(x_i) - p_i| \leq \epsilon$ (Theorem 16). This implies an absolute error $\delta = \epsilon/q$ such that $|\mathcal{D}(x) - \mathcal{D}_\delta(x)| \leq \delta$ for all $x \in S$. The support size is bounded by $|S| \leq 2\tau s + 1$. Using standard tail bounds for discrete Gaussians, the minimum probability satisfies $\mathcal{D}_{\min} = \min_{x \in S} \mathcal{D}(x) \geq$

Algorithm 3: Simultaneous Diophantine Approximation

Input: Real vector $\alpha = (\alpha_0, \alpha_1, \dots, \alpha_{n-1})^\top \in \mathbb{R}^n$, table size n , precision k
Output: Integer vector $\mathbf{SDAT} \in \mathbb{Z}^{n+1}$

- 1 Random $\epsilon \leftarrow (2^{-k/n}, 2^{-k/(n+1)})$
- 2 Initialize empty table $\mathbf{SDAT}$
- 3 $\mathbf{B} = \begin{bmatrix} \epsilon^{-(n+1)} \mathbf{I}_n & -\epsilon^{-(n+1)} \alpha \\ \mathbf{0} & 1 \end{bmatrix}$
- 4 $\mathbf{v} \leftarrow \text{SVP Solver}(\mathcal{L}(\mathbf{B}))$
- 5 Solve $\mathbf{B}\mathbf{x} = \mathbf{v}$ for integer vector $\mathbf{x} = (x_0, \dots, x_n)^\top \in \mathbb{Z}^{n+1}$
- 6 **if** x_i/x_n are k -bit compact approximations for α_i where $0 \leq i \leq n-1$ **then**
- 7 | **return** $\mathbf{SDAT} = \mathbf{x}$
- 8 **end**
- 9 **else**
- 10 | Restart from line 1
- 11 **end**

Algorithm 4: SDA-CDT Offline Phase

Input: CDF $\mathcal{T}_{c,s}$, precision k
Output: Precomputed table $\mathbf{SDAT}[\cdot]$

- 1 Initialize empty vector α
- 2 $\mathcal{T}_{c,s}(\lceil c - \tau s \rceil - 1) = 0$
- 3 **for** $i \leftarrow \lceil c - \tau s \rceil$ **to** $\lceil c + \tau s \rceil$ **do**
- 4 | $\alpha_{i - \lceil c - \tau s \rceil} \leftarrow \mathcal{T}_{c,s}(i) - \mathcal{T}_{c,s}(i - 1)$
- 5 **end**
- 6 $\mathbf{SDAT} \leftarrow \text{SDA}(\alpha, k)$
- 7 **for** $i \leftarrow \lceil c - \tau s \rceil + 1$ **to** $\lceil c + \tau s \rceil$ **do**
- 8 | $\mathbf{SDAT}[i] = \mathbf{SDAT}[i] + \mathbf{SDAT}[i - 1]$
- 9 **end**
- 10 **return** $\mathbf{SDAT}$

Algorithm 5: SDA-CDT Online Phase

Input: Precomputed $\mathbf{SDAT}[\cdot]$
Output: Sample $x \sim \mathcal{D}_{c,s}$

- 1 $q = \mathbf{SDAT}[n]$
- 2 $u \leftarrow \text{UniformBnd}(q)$
- 3 $x = \lceil c - \tau s \rceil - 1$
- 4 **for** $i \leftarrow \lceil c - \tau s \rceil$ **to** $\lceil c + \tau s \rceil$ **do**
- 5 | $x \leftarrow x + \langle u < \mathbf{SDAT}[i] \rangle$
- 6 **end**
- 7 **return** x

$\exp(-\pi\tau^2)/s$. Consequently, the relative error η for any $x \in S$ is bounded by $\eta = \max_{x \in S} \left| \frac{\mathcal{D}(x) - \mathcal{D}_\delta(x)}{\mathcal{D}(x)} \right| \leq \delta/\mathcal{D}_{\min} = \epsilon/(q\mathcal{D}_{\min}) \leq \epsilon s \exp(\pi\tau^2)/q$.

1) *Closeness:* We now present two lemmas that provide upper bounds for SD and RD when approximating CDT.

Lemma 18: Let $\mathcal{D}$ be an ideal distribution and $\mathcal{D}_\delta$ its k -bit compact approximation using SDA. The support S is defined as $\{\lceil c - \tau s \rceil, \lceil c - \tau s \rceil + 1, \dots, \lceil c + \tau s \rceil\}$. We set $\{p_i\}_{i \in S, q}$ to be the integers returned by SDA with $|q\mathcal{D}(i) - p_i| \leq \epsilon$ for $i = 1, \dots, n$.

Then the SD between $\mathcal{D}$ and $\mathcal{D}_\delta$ satisfies:

$$\Delta(\mathcal{D}, \mathcal{D}_\delta) \leq \frac{(2\tau s + 1)\epsilon}{2q}. \quad (28)$$

Proof: By the definition of statistical distance and the properties of SDA:

$$\Delta(\mathcal{D}, \mathcal{D}_\delta) = \frac{1}{2} \sum_{i \in S} \left| \mathcal{D}(i) - \frac{p_i}{q} \right|. \quad (29)$$

Substituting $\mathcal{D}_\delta(i) = p_i/q$ and applying the SDA error bound:

$$\left| \mathcal{D}(i) - \frac{p_i}{q} \right| = \frac{1}{q} |q\mathcal{D}(i) - p_i| \leq \frac{\epsilon}{q}. \quad (30)$$

Let $|S|$ denote the number of elements in the support S . Since S spans from $\lceil c - \tau\sigma \rceil$ to $\lceil c + \tau\sigma \rceil$, we have:

$$|S| = \lceil c + \tau s \rceil - \lceil c - \tau s \rceil + 1 \leq 2\tau s + 1. \quad (31)$$

Summing over all $i \in S$:

$$\Delta(\mathcal{D}, \mathcal{D}_\delta) \leq \frac{1}{2} \cdot |S| \cdot \frac{\epsilon}{q} \leq \frac{(2\tau s + 1)\epsilon}{2q}. \quad (32)$$

For large τs , the $+1$ becomes negligible, yielding the asymptotic bound $\tau s \epsilon / q$. $\square$

Lemma 19: Let $\mathcal{D}$ be an ideal distribution and $\mathcal{D}_\delta$ its k -bit compact approximation using SDA. The support S is defined as $\{\lceil c - \tau s \rceil, \lceil c - \tau s \rceil + 1, \dots, \lceil c + \tau s \rceil\}$. We set $\{p_i\}_{i \in S}, q$ to be the integers returned by SDA with $|q\mathcal{D}(i) - p_i| \leq \epsilon$ for $i = 1, \dots, n$.

Then the RD between $\mathcal{D}$ and $\mathcal{D}_\delta$ satisfies:

$$R_a(\mathcal{D}_\delta \parallel \mathcal{D}) \leq 1 + \frac{a}{a-1} \cdot \frac{\epsilon s \exp(\pi \tau^2)}{q}. \quad (33)$$

Proof: Begin with the relative error expression:

$$\mathcal{D}_\delta(x) = \mathcal{D}(x)(1 + e_x) \quad \text{where} \quad |e_x| \leq \eta. \quad (34)$$

Substitute into Rényi divergence:

$$R_a(\mathcal{D}_\delta \parallel \mathcal{D}) = \left(\sum_{x \in S} \mathcal{D}(x)(1 + e_x)^a \right)^{\frac{1}{a-1}}. \quad (35)$$

Bound the term $(1 + e_x)^a$:

$$(1 + e_x)^a \leq (1 + \eta)^a \quad (\text{since } a > 1 \text{ and } |e_x| \leq \eta). \quad (36)$$

Apply the bound to the summation:

$$\sum_{x \in S} \mathcal{D}(x)(1 + e_x)^a \leq (1 + \eta)^a \sum_{x \in S} \mathcal{D}(x) \leq (1 + \eta)^a. \quad (37)$$

Thus:

$$R_a(\mathcal{D}_\delta \parallel \mathcal{D}) \leq (1 + \eta)^{\frac{a}{a-1}}. \quad (38)$$

For small η (typical in cryptographic settings), use the linear approximation:

$$(1 + \eta)^{\frac{a}{a-1}} \leq 1 + \frac{a}{a-1} \eta \quad (\text{Bernoulli's inequality for } \eta > -1). \quad (39)$$

Determine η from the SDA error:

$$|e_x| = \left| \frac{\mathcal{D}_\delta(x)}{\mathcal{D}(x)} - 1 \right| \leq \frac{\epsilon}{q\mathcal{D}(x)} \leq \frac{\epsilon}{q\mathcal{D}_{\min}}. \quad (40)$$

where $\mathcal{D}_{\min} = \min_{x \in S} \mathcal{D}(x)$. Thus $\eta = \frac{\epsilon}{q\mathcal{D}_{\min}}$.

Apply the minimum probability bound $\mathcal{D}_{\min} \geq \frac{\exp(-\pi\tau^2)}{s}$ for discrete Gaussians:

$$\eta \leq \frac{\epsilon s \exp(\pi\tau^2)}{q}. \quad (41)$$

Combine results:

$$R_a(\mathcal{D}_\delta \parallel \mathcal{D}) \leq 1 + \frac{a}{a-1} \cdot \frac{\epsilon s \exp(\pi\tau^2)}{q}. \quad (42)$$

□

SD-based Analysis.: In this section, the distributions Φ and Φ' denote the cryptographic scheme in the view of the adversary in the approximate (resp. ideal) cases. We assume that a query to the private-key functionality oracle corresponds to m queries to the Gaussian sampling algorithm, with a maximum Gaussian parameter $s : \max_{i=1, \dots, m} s_i$. In accordance with the NIST call for proposals for the post-quantum cryptography standardization, we bound the number of queries from the attacker to the private-key functionality oracle by q_s . Note that in the NIST call for proposals $q_s = 2^{64}$. Any adversary with success probability ϵ' on the scheme implemented with perfect Gaussian sampling has a success probability $\epsilon \leq \epsilon' + \Delta(\Phi, \Phi')$ against the scheme implemented with approximative Gaussian sampling. We assume that the parameters for the ideal scheme are selected to have $\epsilon' \leq 2^{-\lambda-1}$. To ensure security against $m q_s$ samples, the statistical distance per sample must satisfy:

$$\Delta(\mathcal{D}, \mathcal{D}_\delta) \leq \frac{\Delta(\Phi, \Phi')}{m q_s} \leq \frac{1}{2^{\lambda+1} m q_s}. \quad (43)$$

Using Lemma 18 and approximating $2\tau s + 1 \approx 2\tau s$ for large τs :

$$\frac{\tau s \epsilon}{q} \leq \frac{1}{2^{\lambda+1} m q_s}. \quad (44)$$

Rearranging gives the precision requirement:

$$\frac{\epsilon}{q} \leq \frac{1}{2^{\lambda+1} m q_s \tau s}. \quad (45)$$

RD-based Analysis.: Given the SDA-CDT construction, we derive a tight upper bound for the Rényi divergence between the ideal discrete Gaussian distribution $\mathcal{D}$ and its approximation $\mathcal{D}_\delta$. Let Φ and Φ' denote the cryptographic scheme in the view of the adversary under the approximate and ideal distributions, respectively. An adversary makes q_s queries to the secret-key functionality oracle, with each query involving m calls to the Gaussian sampler, resulting in $m \cdot q_s$ total samples.

For λ -bit security, we assume the ideal scheme achieves $(\lambda+1)$ -bit security with success probability $\epsilon' \leq 2^{-\lambda-1}$. The success probability ϵ under $\mathcal{D}_\delta$ must satisfy $\epsilon \leq 2^{-\lambda}$. By the probability preservation property of the Rényi divergence:

$$\epsilon' \geq \epsilon^{a/(a-1)} / R_a(\mathcal{D}_\delta^{m \cdot q_s} \parallel \mathcal{D}^{m \cdot q_s}). \quad (46)$$

Using the multiplicative property $R_a(\mathcal{D}_\delta^{m \cdot q_s} \parallel \mathcal{D}^{m \cdot q_s}) = [R_a(\mathcal{D}_\delta \parallel \mathcal{D})]^{m \cdot q_s}$ and Lemma 19:

$$R_a(\mathcal{D}_\delta \parallel \mathcal{D}) \leq 1 + \frac{a}{a-1} \cdot \frac{\epsilon s \exp(\pi\tau^2)}{q}, \quad (47)$$

we set $a = 2\lambda$ to obtain:

$$2^{-\lambda-1} \geq (2^{-\lambda})^{2\lambda/(2\lambda-1)} / \left[1 + \frac{2\lambda}{2\lambda-1} \cdot \frac{\epsilon s \exp(\pi\tau^2)}{q} \right]^{m \cdot q_s}. \quad (48)$$

For small approximation errors, this simplifies to:

$$\left[1 + \frac{2\lambda}{2\lambda - 1} \cdot \frac{\epsilon s \exp(\pi\tau^2)}{q}\right]^{m \cdot q_s} \leq \sqrt{2}. \quad (49)$$

Using the approximation $\ln(1 + u) \approx u$ for small u :

$$mq_s \cdot \frac{2\lambda}{2\lambda - 1} \cdot \frac{\epsilon s \exp(\pi\tau^2)}{q} \leq \frac{\ln 2}{2}. \quad (50)$$

Solving for ϵ/q yields:

$$\frac{\epsilon}{q} \leq \frac{(2\lambda - 1) \ln 2}{4\lambda m q_s s \exp(\pi\tau^2)} \approx \frac{\ln 2}{2m q_s s \exp(\pi\tau^2)}. \quad (51)$$

Comparison between SD-based Analysis and RD-based Analysis: In evaluating the precision requirements for cryptographic implementations, we contrast the bounds derived from the statistical distance (SD)-based analysis with those from the Rényi divergence (RD)-based analysis. For a security parameter $\lambda = 256$ and a maximum number of queries $q_s = 2^{64}$, the SD-based bound, as established in Lemma 18, requires that the approximation error satisfies

$$\frac{\epsilon}{q} \leq \frac{1}{2^{321} m \tau s}, \quad (52)$$

where m denotes the number of Gaussian sampler calls per query, τ is the tailcut parameter, and s is the Gaussian parameter. This stringent bound necessitates extremely high precision in the approximation.

In contrast, the RD-based analysis, as detailed in Lemma 19 and the subsequent derivation, yields the bound:

$$\frac{\epsilon}{q} \leq \frac{511 \cdot \ln 2}{1024 \cdot m \cdot 2^{64} \cdot s \cdot \exp(\pi\tau^2)} \approx \frac{\ln 2}{2^{65} m s \exp(\pi\tau^2)}. \quad (53)$$

Crucially, this bound scales polynomially in λ rather than exponentially. Although the denominator contains the exponential term $\exp(\pi\tau^2)$, the tail-cut parameter τ is typically chosen as a small fixed constant in lattice-based cryptography. Consequently, for large security parameters λ , the RD-based approach yields significantly more relaxed precision requirements compared to the SD-based analysis. This advantage is particularly pronounced in lattice-based cryptographic primitives requiring high security levels.

2) *Constant-Time Execution:* Given a probabilistic or deterministic algorithm, we consider it to be constant time if its execution time is independent of the sensitive part of its input and output [23]. Rather than ensuring that each operation concludes within a constant interval, the focus is on safeguarding sensitive information from being recovered by timing-based side channel attacks. It is imperative to note that potentially sensitive information encompasses the inputs c and s of the discrete Gaussian, in addition to the yielded samples. The subsequent study will examine which part of SDA-CDT is constant-time and which part is not.

Our focus is on determining whether the individual steps of the online phase are independent from the inputs and outputs. The online phase of the SDA-CDT is a relatively brief process which can be divided into the following steps. Firstly, random integers bounded by integer q are to be generated. Secondly, the **SDAT** is to be traversed. Thirdly, a random integer and an entry of **SDAT** are to be compared at every step of the traversal process and accumulated on x . Finally, the value of x is to be returned.

The initial step of generating a random integer u up to integer q is independent of both c and s , as well as the output x . The second step in the **SDAT** traversal process is associated with s , but not with c . It is evident that the magnitude of s directly correlates to the number of traversal steps and the duration of the process. This is particularly evident in the traversal approach employed by Algorithm 5, where the traversal step is fixed to $2\tau s$. It is important to note that this approach does not come to an end immediately when the smallest i such that $\mathbf{SDAT}[i] > u$ has

been identified. Instead, it comes to an end after the entire table has been traversed. This makes it independent of the output x . However, in the event of an immediate return of the smallest i such that $\mathbf{SDAT}[i] > u$, there exists a possibility of a leak in the value of the sampling result. The selection of an implementation is contingent upon the user's specific requirements. In addition to the linear search, a binary search is also possible to traverse the $\mathbf{SDAT}$ [32].

In Table I, we listed all types of operations in the SDA-CDT sampler and we illustrated the correlation between the execution time and certain input and output of the sampler. The results are categorized as either relevant or irrelevant.

TABLE I: SDA-CDT online phase: Whether the execution time is relevant to (possibly) sensitive information; relevant: ✓; irrelevant: ✗.

operation	positions in Algorithm 5	center c	parameter s	output samples x
taking uniformly randomized u	line 2	✗	✗	✗
table look-up	line 4-6	✗	✓	✗

V. APPLICATION

To evaluate the practical performance of our SDA-CDT construction, we focus on two representative and widely studied lattice-based cryptographic schemes: Frodo [22] and Falcon [33]. Frodo is a post-quantum key encapsulation mechanism that relies on standard lattice assumptions without structured algebraic constructs, and it employs discrete Gaussian sampling with small support. Falcon, in contrast, is a signature scheme based on NTRU lattices that leverages CDT as a base sampler. These two schemes were selected because they (1) rely critically on CDT or CDT-like samplers in their core operations, (2) represent the use cases of both base samplers (Falcon) and full samplers (Frodo), and (3) they are finalists or recommendations in the NIST Post-Quantum Cryptography Standardization process. Hence, they provide ideal benchmarks to demonstrate the performance impact of our SDA-CDT design in realistic cryptographic contexts.

We give a summary of the parameter settings for Falcon and Frodo in Table II. The implementation was done on a PC with CPU of Intel I7-10700K@3.8GHZ with hyperthreading and Turbo disabled. The compiler was g++ 11.4.0 with -O3 -march=native. To support reproducible research, the implementation code is publicly available at github.com/Make1205/SDA_DGS.

TABLE II: Parameter Configurations for Standardized Schemes

Scheme	Variant	σ	Support	Precision (bits)
Frodo	640	2.8	$\{-12, \dots, 12\}$	15
	976	2.3	$\{-10, \dots, 10\}$	15
	1344	1.4	$\{-6, \dots, 6\}$	15
Falcon	-	1.8205	$\{0, \dots, 18\}$	72

1) *Falcon*: We begin with a comparison with the CDT base sampler of Falcon. Falcon is one of the 3 post-quantum signatures selected by NIST for standardization as a summary of 3rd round competition. An important step in the Falcon is the DGS over integers, which employs a Bernoulli-type hybrid sampler, with the base sampler using a CDT. In order to enhance the implementation process, Falcon provides a portable version that utilises fixed precision arithmetic, thereby eliminating all instances of floating points and specialised commands. The setting given in [19], [23] is as follows.

TABLE III: Comparison of Falcon's Probability and SDA Probability.

Ideal Probability	Falcon Probability $\mathcal{P}$ ($/2^{72}$)	SDA Probability $\mathcal{P}_S(/q)$
$\mathcal{D}_{\mathbb{Z}^+,s_0}(0)$	1 697 680 241 746 640 300 030	1 688 501 996 594 986 825 177
$\mathcal{D}_{\mathbb{Z}^+,s_0}(1)$	1 459 943 456 642 912 959 616	1 452 050 498 579 835 817 843
$\mathcal{D}_{\mathbb{Z}^+,s_0}(2)$	928 488 355 018 011 056 515	923 468 626 606 703 851 434
$\mathcal{D}_{\mathbb{Z}^+,s_0}(3)$	436 693 944 817 054 414 619	434 333 026 675 219 970 840
$\mathcal{D}_{\mathbb{Z}^+,s_0}(4)$	151 893 140 790 369 201 013	151 071 954 062 299 508 375
$\mathcal{D}_{\mathbb{Z}^+,s_0}(5)$	39 071 441 848 292 237 840	38 860 208 152 514 969 118
$\mathcal{D}_{\mathbb{Z}^+,s_0}(6)$	7 432 604 049 020 375 675	7 392 420 826 998 016 933
$\mathcal{D}_{\mathbb{Z}^+,s_0}(7)$	1 045 641 569 992 574 730	1 039 988 470 878 764 240
$\mathcal{D}_{\mathbb{Z}^+,s_0}(8)$	108 788 995 549 429 682	108 200 844 703 115 308
$\mathcal{D}_{\mathbb{Z}^+,s_0}(9)$	8 370 422 445 201 343	8 325 169 053 345 920
$\mathcal{D}_{\mathbb{Z}^+,s_0}(10)$	476 288 472 308 334	473 713 492 489 252
$\mathcal{D}_{\mathbb{Z}^+,s_0}(11)$	20 042 553 305 308	19 934 196 346 691
$\mathcal{D}_{\mathbb{Z}^+,s_0}(12)$	623 729 532 807	620 357 435 743
$\mathcal{D}_{\mathbb{Z}^+,s_0}(13)$	14 354 889 437	14 277 281 952
$\mathcal{D}_{\mathbb{Z}^+,s_0}(14)$	244 322 621	243 001 729
$\mathcal{D}_{\mathbb{Z}^+,s_0}(15)$	3 075 302	3 058 676
$\mathcal{D}_{\mathbb{Z}^+,s_0}(16)$	28 626	28 472
$\mathcal{D}_{\mathbb{Z}^+,s_0}(17)$	197	196
$\mathcal{D}_{\mathbb{Z}^+,s_0}(18)$	1	1
$\log_2(\text{SD}(\cdot, \mathcal{D}_{\mathbb{Z}^+,s_0}))$	-54.63	-74.60
$\log_2(R_{513}(\cdot, \mathcal{D}_{\mathbb{Z}^+,s_0}) - 1)$	-77.94	-79.48

- 1) Compute the smallest tailcut w such that the Rényi divergence R_a between the ideal distribution $\mathcal{D}_{\mathbb{Z}^+,s_0}$ and its restriction to $\{0, 1, \dots, w\}$ (denoted by $\mathcal{D}_{[w],s_0}$) satisfy $R_a(\mathcal{D}_{[w],s_0}, \mathcal{D}_{\mathbb{Z}^+,s_0}) \leq 1 + 1/(4q_{bs})$ where q_{bs} is the maximum number of calls to base sampler and $q_{bs} \leq 2^{76}$.
- 2) Rounds the CDT of $\mathcal{D}_{[w],s_0}$ with θ bits of absolute precision as below.
 - For $x \geq 1$, the value $\mathcal{D}_{[w],s_0}$ is truncated as $\mathcal{P}(x) = \lfloor 2^\theta \mathcal{D}_{[w],s_0}(x) \rfloor / 2^\theta$.
 - In order to have a probability distribution, let $\mathcal{P}(0) = 1 - \sum_{x=1}^w \mathcal{P}(x)$.

According to [23], for any $a \geq 509$, Falcon's requirement for the base sampler is $R_a(\mathcal{P}, \mathcal{D}_{\mathbb{Z}^+,s_0}) \leq 1 + 2^{-78}$. The parameter selection given in [19], [23] is $s_0 = \sigma_0 \sqrt{2\pi} = 1.8205 \sqrt{2\pi}$, $\theta = 72$ and $w = 18$. By SDA-CDT, we calculated the probability density mass $\mathcal{P}_S$ such that $R_a(\mathcal{P}_S, \mathcal{D}_{\mathbb{Z}^+,s_0}) \approx 1 + 2^{-79.48} \leq 1 + 2^{-78}$. Table III gives the probability mass function of Falcon's CDT and SDA-CDT, and Table IV gives the data stored in Falcon's CDT and SDA-CDT. Moreover, we also observe that the proposed SDA-CDT outperforms Falcon's CDT in accuracy by providing the measure of statistical distance and Rényi divergence as shown in Table III.

2) *Frodo*: Frodo uses CDT as a full sampler to implement DGS. The distribution is symmetrical, so only half of the table needs to be stored. Frodo's CDT sampler is fixed at 15 bits of precision, with another 1 bit to determine the sign. So the total randomness consumption per sample are 16 bits. Frodo has three sets of parameter settings, each with different s and support. We performed SDA on the probability mass function in the three settings and obtained the approximation shown in Table V together with the measure of accuracy by statistical distance and Rényi divergence. Table VI gives the data stored in Frodo's CDT and our SADT. We define $\mathcal{D}_{\mathbb{Z},s}(\pm x) = \mathcal{D}_{\mathbb{Z},s}(x) + \mathcal{D}_{\mathbb{Z},s}(-x)$ and $\mathcal{T}_{\mathbb{Z},s}(\pm x) = \sum_{i=-x}^x \mathcal{D}_{\mathbb{Z},s}(i)$. It can be seen that our SADT gives a more

TABLE IV: Comparison of Falcon’s table and SDAT.

Ideal CDT	Falcon Table	SDA Table
$\mathcal{T}_{\mathbb{Z}^+, s_0}(0)$	1 697 680 241 746 640 300 030	1 688 501 996 594 986 825 177
$\mathcal{T}_{\mathbb{Z}^+, s_0}(1)$	3 157 623 698 389 553 259 646	3 140 552 495 174 822 643 020
$\mathcal{T}_{\mathbb{Z}^+, s_0}(2)$	4 086 112 053 407 564 316 161	4 064 021 121 781 526 494 454
$\mathcal{T}_{\mathbb{Z}^+, s_0}(3)$	4 522 805 998 224 618 730 780	4 498 354 148 456 746 465 294
$\mathcal{T}_{\mathbb{Z}^+, s_0}(4)$	4 674 699 139 014 987 931 793	4 649 426 102 519 045 973 669
$\mathcal{T}_{\mathbb{Z}^+, s_0}(5)$	4 713 770 580 863 280 169 633	4 688 286 310 671 560 942 787
$\mathcal{T}_{\mathbb{Z}^+, s_0}(6)$	4 721 203 184 912 300 545 308	4 695 678 731 498 558 959 720
$\mathcal{T}_{\mathbb{Z}^+, s_0}(7)$	4 722 248 826 482 293 120 038	4 696 718 719 969 437 723 960
$\mathcal{T}_{\mathbb{Z}^+, s_0}(8)$	4 722 357 615 477 842 549 720	4 696 826 920 814 140 839 268
$\mathcal{T}_{\mathbb{Z}^+, s_0}(9)$	4 722 365 985 900 287 751 063	4 696 835 245 983 194 185 188
$\mathcal{T}_{\mathbb{Z}^+, s_0}(10)$	4 722 366 462 188 760 059 397	4 696 835 719 696 686 674 440
$\mathcal{T}_{\mathbb{Z}^+, s_0}(11)$	4 722 366 482 231 313 364 705	4 696 835 739 630 883 021 131
$\mathcal{T}_{\mathbb{Z}^+, s_0}(12)$	4 722 366 482 855 042 897 512	4 696 835 740 251 240 456 874
$\mathcal{T}_{\mathbb{Z}^+, s_0}(13)$	4 722 366 482 869 397 786 949	4 696 835 740 265 517 738 826
$\mathcal{T}_{\mathbb{Z}^+, s_0}(14)$	4 722 366 482 869 642 109 570	4 696 835 740 265 760 740 555
$\mathcal{T}_{\mathbb{Z}^+, s_0}(15)$	4 722 366 482 869 645 184 872	4 696 835 740 265 763 799 231
$\mathcal{T}_{\mathbb{Z}^+, s_0}(16)$	4 722 366 482 869 645 213 498	4 696 835 740 265 763 827 703
$\mathcal{T}_{\mathbb{Z}^+, s_0}(17)$	4 722 366 482 869 645 213 695	4 696 835 740 265 763 827 899
$\mathcal{T}_{\mathbb{Z}^+, s_0}(18)$	4 722 366 482 869 645 213 696	4 696 835 740 265 763 827 900

compact representation of the ideal distribution $\mathcal{D}_{\mathbb{Z}, s}$ with an improved accuracy.

3) *Summary of Advantages:* The approximation of the d probability values was achieved by employing the SDA (SVP Solver based on infinity norm). However, it is evident that the final probability value in the CDT table is 1, indicating that the last stored integer value is the denominator q . To illustrate this, consider the Falcon Table IV that has been enhanced. In this table, the final element is 4696835740265763827900, consequently establishing that the denominator q is equivalent to 4696835740265763827900. Frodo’s improvement follows the same principle.

The values of $\mathcal{D}_{\mathbb{Z}, s}(\pm 12)$ in Frodo-640 and $\mathcal{D}_{\mathbb{Z}, s}(\pm 10)$ in Frodo-976 obtained from SDA are both 0. However, the support set changes without sacrificing the Rényi divergence and statistical distance. In this case, the values of $\mathcal{T}_{\mathbb{Z}, s}(\pm 12)$ in Frodo-640 and $\mathcal{T}_{\mathbb{Z}, s}(\pm 10)$ in Frodo-976 are no longer stored, which saves space to some extent.

As shown in Table VII, we observe that our SDA-CDT sampler provides notable enhancement in speed, memory consumption and randomness consumption compared with a classical CDT.

In Frodo-1344, SDA-CDT improves sampling speed by 19.97% (from 12.10 to 14.51 million samples per second), demonstrating the highest acceleration across all Frodo variants. In Frodo-976, memory consumption is reduced by 21.60% (from 162 bits to 127 bits), and randomness per sample drops by 12.5% (from 16 bits to 14 bits). In Falcon, SDA-CDT still boosts speed by 8.08% while maintaining constant randomness usage and almost identical memory cost (-0.07%), showing its suitability for high-precision lattice signatures.

The gain is achieved by a more compact representation of the probability mass function derived by SDA. The reason for less memory cost is obvious. The speed is mainly influenced by two factors: generating random integers

TABLE V: Comparison of Frodo's Probability and SDA Probability.

Ideal Probability	Frodo640 ($/2^{15}$)	SDAT($/14534$)	Frodo976($/2^{15}$)	SDAT($/5966$)	Frodo1344($/2^{15}$)	SDAT($/16522$)
$\mathcal{D}_{\mathbb{Z},s}(0)$	4643	2071	5638	1035	9142	4708
$\mathcal{D}_{\mathbb{Z},s}(\pm 1)$	8720	3886	10277	1883	14320	7296
$\mathcal{D}_{\mathbb{Z},s}(\pm 2)$	7216	3209	7774	1418	6876	3394
$\mathcal{D}_{\mathbb{Z},s}(\pm 3)$	5264	2333	4882	884	2024	948
$\mathcal{D}_{\mathbb{Z},s}(\pm 4)$	3384	1493	2545	456	364	159
$\mathcal{D}_{\mathbb{Z},s}(\pm 5)$	1918	841	1101	195	40	16
$\mathcal{D}_{\mathbb{Z},s}(\pm 6)$	958	417	396	69	2	1
$\mathcal{D}_{\mathbb{Z},s}(\pm 7)$	422	182	118	20	-	-
$\mathcal{D}_{\mathbb{Z},s}(\pm 8)$	164	70	29	5	-	-
$\mathcal{D}_{\mathbb{Z},s}(\pm 9)$	56	24	6	1	-	-
$\mathcal{D}_{\mathbb{Z},s}(\pm 10)$	17	7	1	0	-	-
$\mathcal{D}_{\mathbb{Z},s}(\pm 11)$	4	1	-	-	-	-
$\mathcal{D}_{\mathbb{Z},s}(\pm 12)$	1	0	-	-	-	-
σ	2.8		2.3		1.4	
$\log_2(\text{SD}(\cdot, \mathcal{D}_{\mathbb{Z},s}))$	-8.59	-13.32	-8.03	-13.31	-6.57	-17.70
$\log_2(R_a(\cdot, \mathcal{D}_{\mathbb{Z},s}) - 1)$	-4.96	-7.59	-3.32	-6.53	-1.98	-5.33
a	200		500		1000	

TABLE VI: Comparison of Frodo's table and SDAT.

Ideal CDT	Frodo640 ($/2^{15}$)	SDAT($/14534$)	Frodo976($/2^{15}$)	SDAT($/5966$)	Frodo1344($/2^{15}$)	SDAT($/16522$)
$\mathcal{T}_{\mathbb{Z},s}(0)$	4643	2071	5638	1035	9142	4708
$\mathcal{T}_{\mathbb{Z},s}(\pm 1)$	13363	5957	15915	2918	23462	12004
$\mathcal{T}_{\mathbb{Z},s}(\pm 2)$	20579	9166	23689	4336	30338	15398
$\mathcal{T}_{\mathbb{Z},s}(\pm 3)$	25843	11499	28571	5220	32361	16346
$\mathcal{T}_{\mathbb{Z},s}(\pm 4)$	29227	12992	31116	5676	32725	16505
$\mathcal{T}_{\mathbb{Z},s}(\pm 5)$	31145	13833	32217	5871	32765	16521
$\mathcal{T}_{\mathbb{Z},s}(\pm 6)$	32103	14250	32613	5940	32767	16522
$\mathcal{T}_{\mathbb{Z},s}(\pm 7)$	32525	14432	32731	5960	-	-
$\mathcal{T}_{\mathbb{Z},s}(\pm 8)$	32689	14502	32760	5965	-	-
$\mathcal{T}_{\mathbb{Z},s}(\pm 9)$	32745	14526	32766	5966	-	-
$\mathcal{T}_{\mathbb{Z},s}(\pm 10)$	32762	14533	32767	-	-	-
$\mathcal{T}_{\mathbb{Z},s}(\pm 11)$	32766	14534	-	-	-	-
$\mathcal{T}_{\mathbb{Z},s}(\pm 12)$	32767	-	-	-	-	-

(i.e. line 2 of Algorithm 5) and comparing two integers (i.e. line 5 of Algorithm 5). As shown in Tables IV and VI, the proposed method gives a compact representation of the desired distribution with a smaller denominator q as well as smaller numerators than a classical CDT. This accounts for the improvement of speed and randomness cost.

VI. CONCLUSION

This paper has presented a new approach to discrete Gaussian sampling through the development of SDA-CDT, a compact and constant-time sampler based on simultaneous Diophantine approximation. Our work bridges an important gap between theoretical lattice-based cryptography and practical implementation constraints by addressing the inefficient data representation of CDT samplers while maintaining rigorous security guarantees. From an implementation perspective, SDA-CDT offers three notable advantages over existing approaches. Firstly, the more compact data representation benefits the speed, memory and randomness cost. Secondly, the shared denominator representation naturally supports constant-time execution, crucial for side-channel resistance. Thirdly, the compatibility with Rényi divergence analysis allows for optimized parameter selection, particularly beneficial for schemes with bounded query complexity.

TABLE VII: Performance Comparison: SDA-CDT vs Classical CDT

Scheme	Speed ($10^6/s$)		Memory (bits)		Randomness (bits/sample)		Improvement (%)		
	CDT	SDA-CDT	CDT	SDA-CDT	CDT	SDA-CDT	Speed	Memory	Randomness
Frodo-640	10.51	11.30	192	165	16	15	+7.49	-14.06	-6.25
Frodo-976	11.55	12.80	162	127	16	14	+10.88	-21.60	-12.50
Frodo-1344	12.10	14.51	104	100	16	16	+19.97	-3.85	0
Falcon	9.40	10.17	1368	1367	72	72	+8.08	-0.07	0

The techniques developed in this work open several interesting avenues for future research. Adapting the SDA-CDT sampler to non-centered distributions could further enhance its utility in advanced cryptographic constructions, particularly those requiring efficient sampling from shifted or skewed distributions. The development of hardware-aware optimizations—such as specialized arithmetic units for rational number operations—could yield additional performance gains in resource-constrained environments. Further, exploring the integration of our techniques with other sampling methods, such as combining SDA-CDT with convolution-based approaches, may lead to new trade-offs between efficiency and security for schemes requiring variable parameter sampling. These extensions would further demonstrate the versatility of SDA as a tool for optimizing lattice-based cryptographic implementations.

APPENDIX A

ENUMERATION ALGORITHM FOR UPPER-TRIANGULAR BASIS

We present an adjusted enumeration algorithm for solving ℓ_∞ -norm SVP under the given upper-triangular lattice basis structure:

$$\mathbf{B} = \begin{bmatrix} \epsilon^{-(n+1)} \mathbf{I}_n & -\epsilon^{-(n+1)} \boldsymbol{\alpha} \\ \mathbf{0} & 1 \end{bmatrix}, \quad (54)$$

where $0 < \epsilon < 1$.

Note that we can use the solution of truncation to set the initial radius of the enumeration. If it is not the optimal solution, the algorithm output must be better. Denote the fixed-point truncation of $\boldsymbol{\alpha}$ as $(\frac{p'_1}{2^k}, \dots, \frac{p'_n}{2^k}) = (\frac{\lfloor 2^k \alpha_1 \rfloor}{2^k}, \dots, \frac{\lfloor 2^k \alpha_n \rfloor}{2^k})$. This can be used to construct a short vector of lattice $\mathcal{L}$:

$$\mathbf{B} \cdot [p'_1, p'_2, \dots, p'_n, 2^k]^\top. \quad (55)$$

Letting $\epsilon = 2^{-k/n}$, this initial searching radius ρ of the enumeration algorithm is

$$\rho = \left\| \begin{bmatrix} 2^{k(n+1)/n} \mathbf{I}_n & -2^{k(n+1)/n} \boldsymbol{\alpha} \\ \mathbf{0} & 1 \end{bmatrix} \cdot [\lfloor 2^k \alpha_1 \rfloor, \lfloor 2^k \alpha_2 \rfloor, \dots, \lfloor 2^k \alpha_n \rfloor, 2^k]^\top \right\|_\infty.$$

A. Algorithm Design

The upper-triangular structure simplifies partial sum calculations. For $\mathbf{x} = (x_1, \dots, x_{n+1})^\top \in \mathbb{Z}^{n+1}$, define $y_k = (\mathbf{B}\mathbf{x})_k$ as:

$$y_k = \begin{cases} (x_k - \alpha_k x_{n+1}) \epsilon^{-(n+1)}, & \text{for } k = 1, \dots, n, \\ x_{n+1}, & \text{for } k = n+1. \end{cases}$$

The ℓ_∞ constraint $\max_{1 \leq k \leq n+1} |y_k| \leq \rho$ reduces to:

- For $k = n+1$: $|x_{n+1}| \leq \rho \Rightarrow x_{n+1} \in (-\rho, \rho)$.
- For $k = 1, \dots, n$: Given x_{n+1} , each x_k must satisfy

$$x_k \in (\alpha_k x_{n+1} - \epsilon^{(n+1)} \rho, \alpha_k x_{n+1} + \epsilon^{(n+1)} \rho).$$

To exploit the special structure of the upper-triangular basis, we design a customized enumeration method tailored for the ℓ_∞ -norm. This approach significantly reduces the search space by leveraging the deterministic dependencies among the coordinates induced by the basis. The full procedure is detailed in Algorithm 6, which systematically explores feasible integer vectors $\mathbf{x}$ by enumerating the final coordinate x_{n+1} and recursively bounding the remaining entries using norm constraints. This structure-aware traversal allows aggressive pruning and yields improved enumeration efficiency over generic SVP solvers.

Algorithm 6: Sphere Decoding for SVP (SDSVP)

Input: Lattice basis $\mathbf{B} \in \mathbb{R}^{m \times n}$, search radius *radius*

Output: Integer coefficient vector $\mathbf{x}$ of shortest non-zero lattice vector

```

[m, n] ← size(B)
z = 0min(m,n)                                     ▷ Origin point
SPHDEC_RADIUS = radius
RETV = 0n
x = 0n
NUMX = 0
sphdec_core(z, B, n, 0)                             ▷ Recursive sphere decoding
if NUMX > 0 then                                     ▷ Shortest vector found
  | return RETV
else
  | return 0n                                         ▷ No solution
end

```

B. Complexity Analysis

Lemma 20: Let $\mathcal{L}$ is a lattice and $\rho \in \mathbb{R}_{>0}$. Then $|B_n^{(\infty)}(\rho) \cap \mathcal{L}| \leq \left(1 + \left\lfloor \frac{2\rho}{\lambda_1^{(\infty)}} \right\rfloor\right)^n$.

Proof: Note that for any non-negative integers $i_1, \dots, i_n$, the region

$$[-\rho + i_1 \lambda_1^{(\infty)}, -\rho + (i_1 + 1) \lambda_1^{(\infty)}) \times [-\rho + i_2 \lambda_1^{(\infty)}, -\rho + (i_2 + 1) \lambda_1^{(\infty)}) \times \dots \times [-\rho + i_n \lambda_1^{(\infty)}, -\rho + (i_n + 1) \lambda_1^{(\infty)}) \quad (56)$$

contains at most one lattice point. The values of i_j for any $j \in [n]$ such that this region intersects with $B_n^{(\infty)}(\rho)$ are $\{0, 1, \dots, \left\lfloor \frac{2\rho}{\lambda_1^{(\infty)}} \right\rfloor\}$. The result follows. $\square$

Therefore, the worst-case node count in enumeration is

$$N_\infty(\rho) \leq \left(1 + \left\lfloor \frac{2\rho}{\lambda_1^{(\infty)}} \right\rfloor\right)^n.$$

However, the actual count is often smaller due to:

- **Linear coupling:** Values of x_k are tightly coupled to x_{n+1} via α_k , reducing independent variability.
- **Dynamic pruning:** Empty intervals for x_k eliminate entire subtrees.

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Algorithm 7: Sphere Decoder Core

```

Input: Target  $\mathbf{z}$ , matrix  $\mathbf{B}$ , layer  $layer$ , distance  $dist$ 

 $n \leftarrow \text{columns}(\mathbf{B})$ 
if  $layer < n$  then
   $\mathbf{z}' = \mathbf{z} - \mathbf{B}[:, layer + 1 : n] \cdot \mathbf{x}[layer + 1 : n]$ 
else
   $\mathbf{z}' = \mathbf{z}$ 
end
 $c = \lfloor \mathbf{z}'[layer] / \mathbf{B}[layer, layer] \rfloor$  ▷ Babai point
 $\mathbf{x}[layer] = c$ 
 $d = \|\mathbf{z}'[layer] - \mathbf{B}[layer, layer : n]^\top \mathbf{x}[layer : n]\|^2 + dist$ 
if  $d \leq SPHDEC\_RADIUS$  then
  if  $layer == 1$  then
    if  $d \neq 0$  then
       $RETVAL = \mathbf{x}$  ▷ Update solution
       $SPHDEC\_RADIUS = d$  ▷ Shrink radius
       $NUMX = NUMX + 1$ 
    end
  else
     $\text{sphdec\_core}(\mathbf{z}, \mathbf{B}, layer - 1, d)$  ▷ Recurse
  end
end
 $\delta = 0$ 
while  $d \leq SPHDEC\_RADIUS$  do
   $\delta \leftarrow \delta + 1$  ▷ Expand search
  for  $k = 1$  to  $2$  do
     $c_i = c + (-1)^k \cdot \delta$  ▷ Candidate
     $\mathbf{x}[layer] = c_i$ 
     $d = \|\mathbf{z}'[layer] - \mathbf{B}[layer, layer : n]^\top \mathbf{x}(layer : n)\|^2 + dist$ 
    if  $d \leq SPHDEC\_RADIUS$  then
      if  $layer == 1$  then
        if  $d \neq 0$  then
           $RETVAL = \mathbf{x}$  ▷ Update solution
           $SPHDEC\_RADIUS = d$ 
           $NUMX = NUMX + 1$ 
        end
      else
         $\text{sphdec\_core}(\mathbf{z}, \mathbf{B}, layer - 1, d)$  ▷ Recurse
      end
    end
    else
       $\text{break}$  ▷ Prune branch
    end
  end
end
end

```

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